Gait Control of the Lizard V1 Robot

A Self-Contained Mathematical Reference for the MuJoCo Simulation

Lizard Robot Project

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Preface

This document is a complete, self-contained description of the locomotion controllers used in the **Lizard V1** MuJoCo simulation. It is written so that a reader with first-year calculus and trigonometry — and no prior knowledge of this project — can understand exactly how every joint command is generated.

Each control law is presented as a set of equations, followed by the meaning of every symbol, the reason the equation has the form it does, and the numerical values currently used. Two laws are described:

1. the **Sine law**, a smooth traveling-wave gait that we optimized for fast, straight walking; and
2. the **Hardware law**, a port of the real robot's controller, in which the body holds a **standing "C" posture** that flips side in time with a **diagonal trot** of the legs.

Chapter 1. The Robot and Its Notation

1.1 Degrees of freedom

The robot has a central body with an articulated spine and four single-joint legs, for a total of **seven actuated revolute joints**:

Group	Joints	Axis	Role
Spine	Front, Back, Tail	vertical (yaw)	lateral body bending
Front legs	FR, FL	longitudinal	press / lift the front feet
Rear legs	HR, HL	longitudinal	press / lift the rear feet

Each joint is driven by a position-controlled Dynamixel XL430 servo, whose commanded angle we write as $q_j(t)$ for joint j at time t (in radians).

1.2 Reference frame

We use a right-handed world frame fixed to the ground at the spawn point: $+x$ is forward, $+z$ is up, and $+y$ is to the left. The robot's net progress is measured by the forward displacement of its body centre, $\Delta x = x(T) - x(T_s)$, evaluated after the start-up transient (Section 2.2).

1.3 Index sets and labels

Let the ordered list of spine joints be $S = (\text{Front}, \text{Back}, \text{Tail})$ with $N_s = 3$ members, and assign each spine joint an integer **backbone index**:

$$i(\text{Front}) = 0, i(\text{Back}) = 1, i(\text{Tail}) = 2.$$

The four legs are grouped two ways that we use repeatedly:

$$\text{right} = \{\text{FR}, \text{HR}\}, \text{left} = \{\text{FL}, \text{HL}\},$$

$$\text{front} = \{\text{FR}, \text{FL}\}, \text{hind} = \{\text{HR}, \text{HL}\}.$$

Each leg ℓ also has a **girdle** $g(\ell) \in S$ — the spine joint it is attached near (Front for the front legs, Back for the rear legs).

Chapter 2. Common Control Architecture

Both laws are **open-loop**: the controller is a pure function of time. At every simulation step it computes a target angle $q_j(t)$ for each joint, and the MuJoCo position actuator applies a torque proportional to the tracking error, saturated at the servo stall torque.

2.1 The general form

Every joint command has the same three-part structure,

$$q_j(t) = \theta_j + r(t) w_j(t),$$

where θ_j is the **initial angle** (or *offset*) of joint j — the posture the joint holds before the gait starts and the value about which it oscillates; $w_j(t)$ is the time-varying **oscillation** produced by the chosen control law (Chapters 3–4); and $r(t) \in [0, 1]$ is a dimensionless **start-up ramp** shared by all joints.

2.2 The start-up ramp

To avoid a violent jump when the gait switches on, the oscillation amplitude is faded in with a raised-cosine ramp over a settling time T_s followed by a ramp time T_r:

$$r(t)=\begin{cases} 0, & t \le T_s, \\ \frac{1}{2}\left[1-\cos\left(\pi \frac{t-T_s}{T_r}\right)\right], & T_s < t < T_s + T_r, \\ 1, & t \ge T_s + T_r. \end{cases}$$

During the settling phase ($t \le T_s$) every joint simply holds its offset θ_j, letting the robot settle onto the ground. The ramp is C^1-continuous, so both r and its derivative start at zero and there is no acceleration discontinuity. Default values are $T_s = 1.0$ s and $T_r = 1.5$ s.

Chapter 3. The Sine Law (Traveling Wave)

The sine law treats the body as a continuous medium carrying a sinusoidal traveling wave, with the legs phase-locked to it.

3.1 Spine: a traveling wave

For a spine joint with backbone index i,

$$q_i(t) = \theta_i + r(t)\, b + r(t)\, A_s \sin(2\pi f t + \varphi_i),$$

with the **spatial phase**

$$\varphi_i = \frac{2\pi k c}{N_s - 1} i.$$

The symbols are:

Symbol	Name	Meaning
A_s	spine.amplitude	peak bending angle of a spine joint (rad)
f	gait.frequency	temporal frequency of oscillation (Hz)
k	spine.wavenumber	number of full sine waves along the body
c	gait.wave_speed	spatial phase-propagation multiplier
b	spine.turn_trim	constant yaw bias used to steer (rad)
θ_i	offset	per-joint initial angle (rad)

Why this form. The term $2\pi f t$ makes every joint oscillate at the same frequency f. The spatial term φ_i adds a fixed phase lag between consecutive joints; because i runs from 0 to $N_s - 1$, the head-to-tail phase difference is $\varphi_{N_s-1} - \varphi_0 = 2\pi k c$, i.e. exactly $k c$ full waves span the body. A nonzero lag makes the crest travel from head to tail, which propels the robot; the steering bias b shifts the whole body slightly to one side to trim a curved path into a straight one.

3.2 Legs: antisymmetric sine

Each leg oscillates at the same frequency, locked to the spatial phase of its girdle, with the **left legs exactly opposite the right legs**:

$$q_\ell(t)=\theta_\ell+r(t)\sigma_\ell A_\ell \sin(2\pi f t+\varphi_{g(\ell)}+2\pi\psi),$$

where the **side sign** is

$$\sigma_\ell=\begin{cases}+1, & \ell\in \text{right}=\{\text{FR},\text{HR}\},\\ -1, & \ell\in \text{left}=\{\text{FL},\text{HL}\}.\end{cases}$$

Here A_ℓ is legs.amplitude and ψ is the single leg-versus-body phase offset legs.phase_offset (in cycles, hence the factor 2π). The sign σ_ℓ makes the left and right legs mirror one another, so their lateral reaction forces tend to cancel and the net motion is forward.

Chapter 4. The Hardware Law (Standing C-Wave + Diagonal Trot)

This law reproduces the real robot's controller. The body no longer carries a traveling wave; instead it bends into a single **C** shape that reverses each half-cycle, while the legs run a **duty-cycled diagonal trot** locked to that bending.

4.1 Spine: a standing C-wave

Every spine joint shares **one and the same** sinusoid — there is no per-segment phase lag:

$$q_i(t)=\theta_i+r(t)A_b m_i \sin(2\pi f_\ell t+\lambda).$$

Symbol	Name	Meaning
A_b	body_amp	C-bend amplitude (rad)
f_ℓ	leg_freq	gait frequency; the body bends at the leg frequency (Hz)
λ	body_lock	phase that times the bend against foot contact (rad)
m_i	body_shape	per-joint shape weight, $m_i\in\{+1,-1\}$

Because the time argument $2\pi f_\ell t+\lambda$ is **identical** for all joints, the body is a *standing* wave: at any instant every spine joint is displaced by the same signed factor $\sin(2\pi f_\ell t+\lambda)$, scaled by its shape weight m_i. As that factor swings between $+1$ and -1, the body curls one way, straightens, and curls the other way.

4.2 Why the C requires opposite signs (the C-vs-S rule)

A natural guess is to set all shape weights equal, $m_{\text{Front}}=m_{\text{Back}}=m_{\text{Tail}}=+1$. This produces an **S**, not a **C**. The reason is geometric. The Front section extends in $+x$ from the body centre, while the Back and Tail sections extend in $-x$. All three yaw joints rotate about the same world axis $-\hat{z}$, so a positive joint angle rotates any attached point by the same sense. Under a rotation by $+\alpha$ about $-\hat{z}$, a tip pointing in $+x$ moves toward $-y$, while a tip pointing in $-x$ moves toward $+y$. Hence equal-sign commands send the head and the tail to **opposite** lateral sides (an S). To bend head and tail to the **same** side (a C), the rear joints must take the **opposite** sign of the front joint:

$$m_{\text{Front}}=+1,m_{\text{Back}}=m_{\text{Tail}}=-1.$$

This is a property of the kinematic layout, **not** of the joint-axis polarity (all three axes share the same positive direction).

4.3 Legs: the duty-cycle function

A leg should press its foot down for part of the cycle (stance) and lift it for the rest (swing). This is captured by a **duty function** of the cycle fraction $u\in[0,1)$ and the stance fraction $d=$ duty:

$$D(u;d)=\begin{cases}\sin\left(\frac{\pi u}{d}\right), & 0\le u<d\,(\text{stance: foot down}),\\ 0, & d\le u<1\,(\text{swing: foot neutral}).\end{cases}$$

During stance the foot makes one smooth down-and-up press (the half-sine rises to 1 at $u=d/2$ and returns to 0); during swing it rests at the neutral posture. The fraction d is the **duty factor**: $d=0.5$ means the foot is down for half of each stride.

4.4 Legs: the diagonal trot

Each leg's place in the cycle is set by adding a phase $\phi_{lr}=$ phi_leg for being on the right and a phase $\phi_{fb}=$ phi_lat for being at the rear:

$$\phi_\ell(t)=2\pi f_\ell t+[\ell\in\text{right}]\phi_{lr}+[\ell\in\text{hind}]\phi_{fb},$$

where $[\cdot]$ equals 1 when the condition holds and 0 otherwise. The cycle fraction and the final leg command are then

$$u_\ell(t)=\left\langle\frac{\phi_\ell(t)}{2\pi}\right\rangle,q_\ell(t)=\theta_\ell+r(t)e_\ell A_\ell D(u_\ell(t);d),$$

where $\langle\cdot\rangle$ denotes the fractional part, $A_\ell=$ leg_amp, and $e_\ell=$ down_sign $\in\{+1,-1\}$ is the sign that drives each particular foot **downward** (it differs between legs because of mirrored mounting). With the canonical choice $\phi_{lr}=\phi_{fb}=\pi$, the four legs split into the two diagonal pairs of a **trot**:

Leg	Added phase	Contact window
FL (left, front)	0	first half
HR (right,	$\pi+\pi\equiv 0$	first half

hind)		
FR (right, front)	π	second half
HL (left, hind)	π	second half

Thus the pair $\{\mathrm{FL}, \mathrm{HR}\}$ presses together, then $\{\mathrm{FR}, \mathrm{HL}\}$ presses together — the diagonal gait of a walking lizard.

4.5 Locking the bend to the contact

The body phase λ (body_lock) is chosen so the C-bend and the foot contacts are synchronized. At the tuned value the rule is: when the body bends **left**, the pair $\{\mathrm{FL}, \mathrm{HR}\}$ is in contact; when it bends **right**, the pair $\{\mathrm{FR}, \mathrm{HL}\}$ is in contact. In words, **the body always bends toward the side whose front foot is planted.** Shifting λ by π swaps the two correspondences (a half-cycle offset), which is exactly how the relationship was corrected during tuning.

Chapter 5. Optimization of the Gait

The continuous parameters of either law are tuned by simulation-in-the-loop search. We run many short head-less rollouts and score each by the body's **forward progress**.

5.1 Objectives

For raw distance we maximize the forward displacement after the settling time,

$$J_{\mathrm{fwd}} = x(T) - x(T_s).$$

For a **straight** fast gait we penalize lateral drift Δy and net heading change $\Delta\psi$ (final yaw minus initial yaw):

$$J_{\mathrm{straight}} = \Delta x - w_y |\Delta y| - w_\psi |\Delta\psi|,$$

with weights $w_y = 2.0\, m^{-1}$ and $w_\psi = 0.4\, rad^{-1}$ used in this project. A rollout that diverges (body height greater than 1 m) is assigned a large negative score so the optimizer avoids unstable regions.

5.2 Search

The free parameters (amplitudes, phase offsets, duty factor, initial postures, and — when allowed — frequency) are searched with a Tree-structured Parzen Estimator (Optuna). Each study is persisted to disk and is therefore resumable. When the gait *identity* must be preserved (the C shape and the bend-to-front-foot rule), the structural quantities m_i, ϕ_{lr}, ϕ_{fb}, and e_ℓ are held fixed and λ is restricted to a neighbourhood of π.

Appendix A. Symbol Table

Symbol	YAML key	Units	Description
q_j	—	rad	commanded angle of joint j
θ_j	offset.*	rad	per-joint initial angle / oscillation centre
$r(t)$	—	—	start-up ramp, from 0 to 1
T_s, T_r	settle_time, ramp_time	s	settle and ramp durations
f	gait.frequency	Hz	sine-law temporal frequency
c	gait.wave_speed	—	sine-law spatial-propagation multiplier
A_s	spine.amplitude	rad	sine spine amplitude
k	spine.wavenumber	—	waves along the body
b	spine.turn_trim	rad	steering bias
A_ℓ (sine)	legs.amplitude	rad	sine leg amplitude
ψ	legs.phase_offset	cycles	sine leg-vs-body phase
σ_ℓ	legs.right, legs.left	—	leg side sign ± 1
A_b	hardware.body_amp	rad	C-bend amplitude
λ	hardware.body_lock	rad	body-vs-leg phase
m_i	hardware.body_shape	—	spine shape weights ($+1$ front, -1 rear gives a C)
A_ℓ (hw)	hardware.leg_amp	rad	foot-down amplitude
f_ℓ	hardware.leg_freq	Hz	hardware body and leg frequency
d	hardware.duty	—	stance fraction
ϕ_{lr}	hardware.phi_leg	rad	left/right leg phase
ϕ_{fb}	hardware.phi_lat	rad	front/back leg phase
e_ℓ	hardware.down_sign	—	foot-down sign ± 1

Appendix B. Reference Parameter Set

The hardware-law values currently in configs/gait.yaml:

Parameter	Value
f_ℓ (leg_freq)	0.2 Hz
A_b (body_amp)	0.8 rad
λ (body_lock)	2.234 rad
m_i (body_shape)	$(+1, -1, -1)$ for (Front, Back, Tail)

A_ℓ (leg_amp)	0.564 rad
d (duty)	0.491
ϕ_{lr}, ϕ_{fb}	π, π
e_ℓ (down_sign)	$(-1, -1, +1, -1)$ for (FR, FL, HR, HL)
θ_ℓ (offset, legs)	$\mp$ 0.165 rad
T_s, T_r	1.0 s, 1.5 s